Homework 2

CEE 530

Prestressed Concrete

Due 2/11/2026

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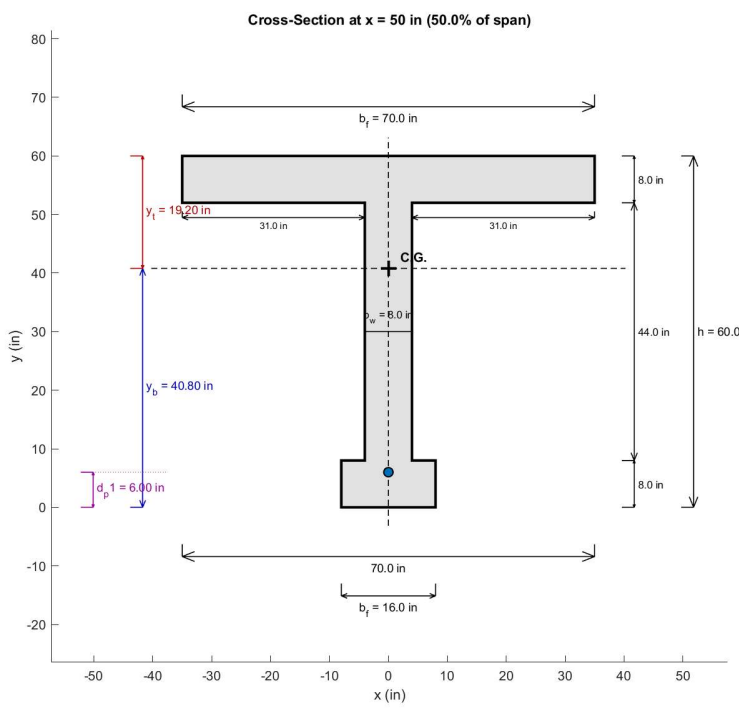
Homework Question and Given Information

In continuation with Homework 1, the prestressing cable is parabolic. Find the stress at 0.5L, 0.75L and 1L and plot bending moment diagram

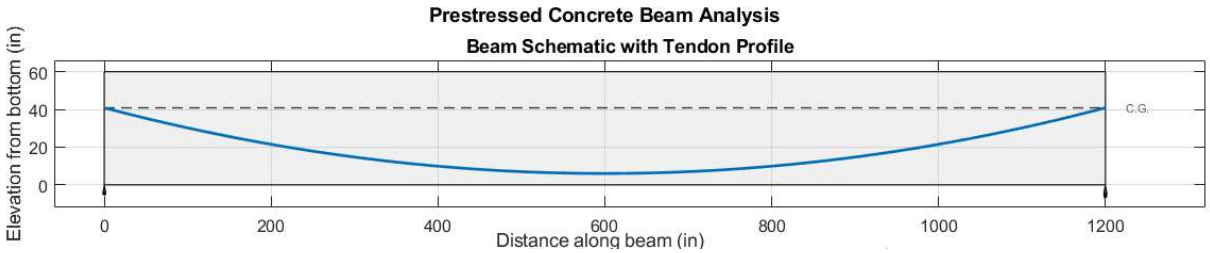
Calculations from HW 1	
Area	1040 in²
y_b	40.8 in
y_t	19.2 in
I	404,241.07 in⁴
S_{top}	21,054 in³
S_{bot}	9,908 in³
E	4.41 x 10⁶ psi



Geometry Input;

Beam Span, L=100ft = 1200 in

Tendon profile, $y = ax^2$



Calculate the variable a

$a(L/2)^2 = (Y_b\text{-cover})$

$a = \frac{4(Yb - cover)}{L^2}$

$a = \frac{4(40.8 - 6)}{1200^2} = 9.67 \times 10^{-5}$

$Y = 9.67 \times 10^{-5} x^2$

At end support, x = L/2 = 600in, from midspan

$Y = 9.67 \times 10^{-5} \times 600^2 = 34.8$ in

e = (40.8 - 6) - 34.8 = 0 in

e = 0 in

At midway of end support and beam midspan, x = 0.5(L/2) = 300in, from midspan

$Y = 9.67 \times 10^{-5} \times 300^2 = 8.7$ in

e = (40.8 - 6) - 8.7 in

e = 26.1 in

At middle of beam, x = 0(L/2) = 0 in, from midspan

$Y = 9.67 \times 10^{-5} \times 0^2 = 0$ in

e = (40.8 - 6) - 0 in

e = 34.8 in

Bending moment (BM) due to self weight (W)

Concrete weight, W = 150 lb/ft^3 = 0.086806 lb/in^3

Uniform load from self-weight, w = concrete unit weight x A

w = 0.08606 x 1040 = 90.28 lb/in

Moment due to load equation, M(x) = 0.5wLx - 0.5wx^2

i) Moment at x=L and x= 0:

M(L) = 0 lb.in

ii) Moment at x=0.75L and x=0.25L:

M(0.25L) = (0.5 x 0.25L x Lx 90.28) - (0.5 x 0.25^2 x L^2 x 90.28)

M(0.75L) = (0.5 x 0.75L x Lx 90.28) - (0.5 x 0.75^2 x L^2 x 90.28)

M = 8.464L^2 = 12,188,160 lb.in

iii) Moment at x=0.5L:

$$M(0.5L) = \frac{w}{2}(L(0.5L) - (0.5L)^2) = \frac{w}{2}(0.5L^2 - 0.25L^2) = \frac{w}{2}(0.25L^2) = \frac{wL^2}{8}$$

M(0.5L) = (90.28L^2)/8

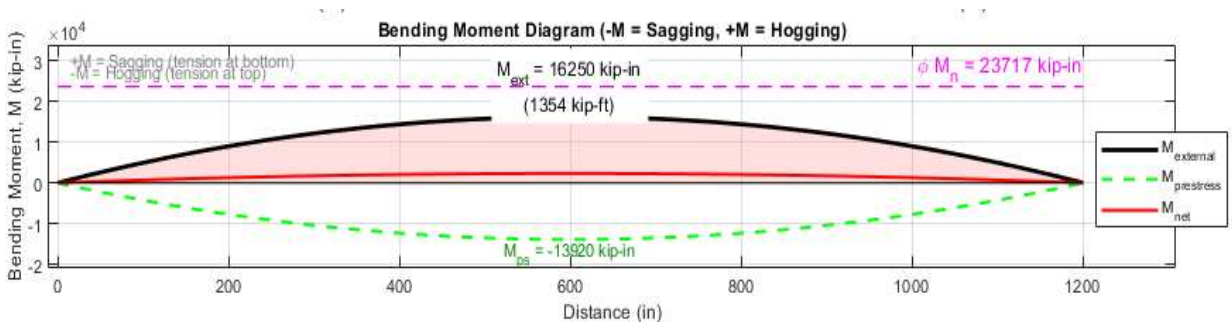
M = 11.285L^2 = 16,250,400 lb.in

Including the moment due to prestressing

M(x) = 0.5wLx - 0.5wx^2 + P*e(x)

Summary of Moments

Location	Moment due to self weight (lb.in)	Moment due to prestressing force (lb.in)	Combine Bending moment (lb.in)
X = 0.5L	16,250,400	-13,920,000	2,330,400
X = 0.75L	12,188,160	-10,440,000	1,748,160
X = L	0	0	0



Stress in Beam

i) Stress due to prestressing force, P

Given :

P = 400 kips

A = 1040 in^3

I = 404241 in^4

Yt = 19.2 in

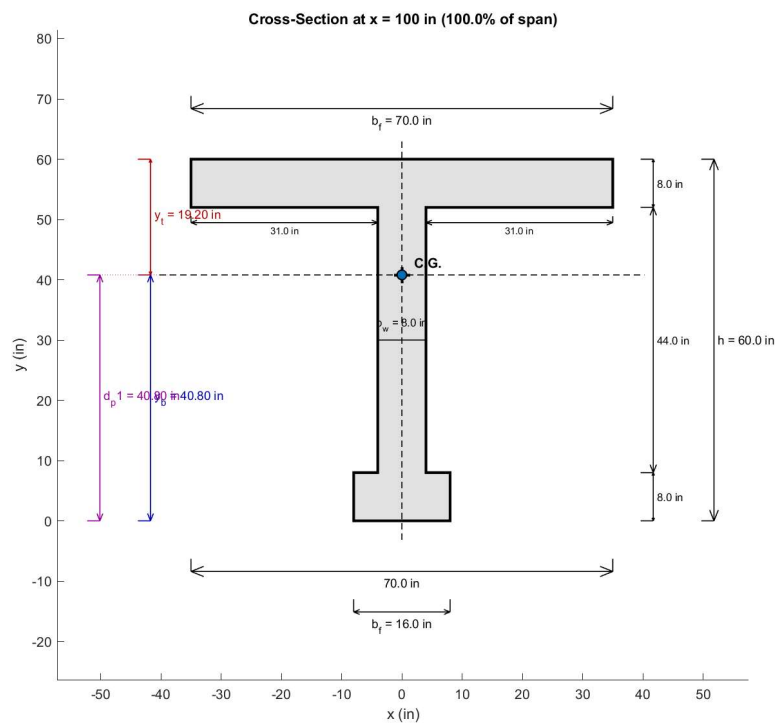
Yb = 40.8 in

Assuming the tendon angle is so small such that the vertical component of stress is negligible.

$\sigma = -\frac{P}{A} = -\frac{400}{1040} = -0.385$ ksi across all sections.

ii) Stress from prestress moment

At end of support, x = L and x = 0



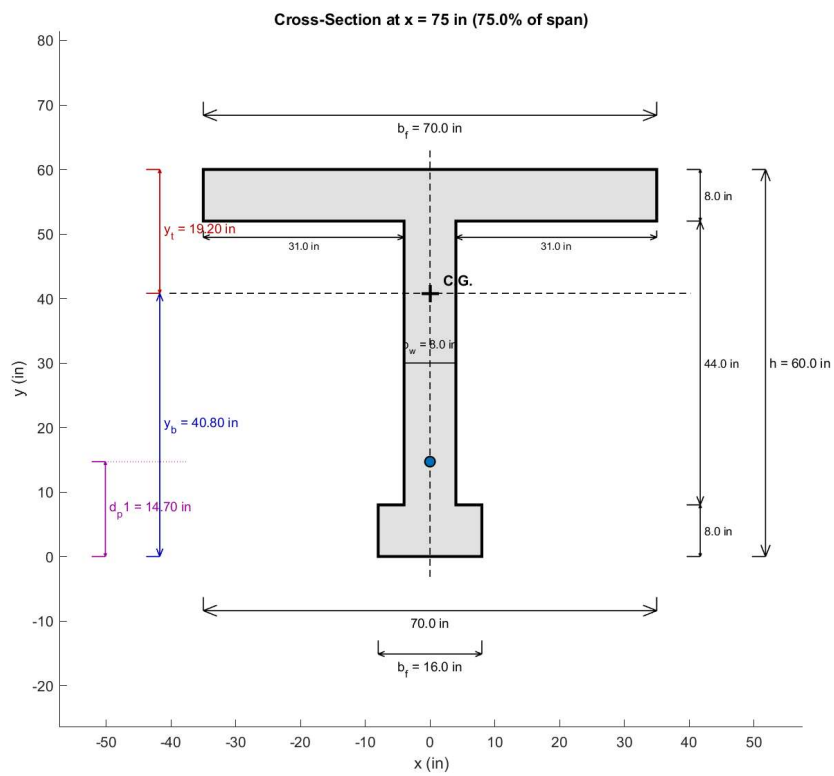
- Top fiber

$$\frac{Pey}{I} = \frac{400(0)(19.2)}{404241.07} = 0 \text{ ksi}$$

- Bottom Fiber

$$\frac{Pey}{I} = \frac{400(0)(40.8)}{404241.07} = 0 \text{ ksi}$$

At x = 0.75L and x = 0.25L,



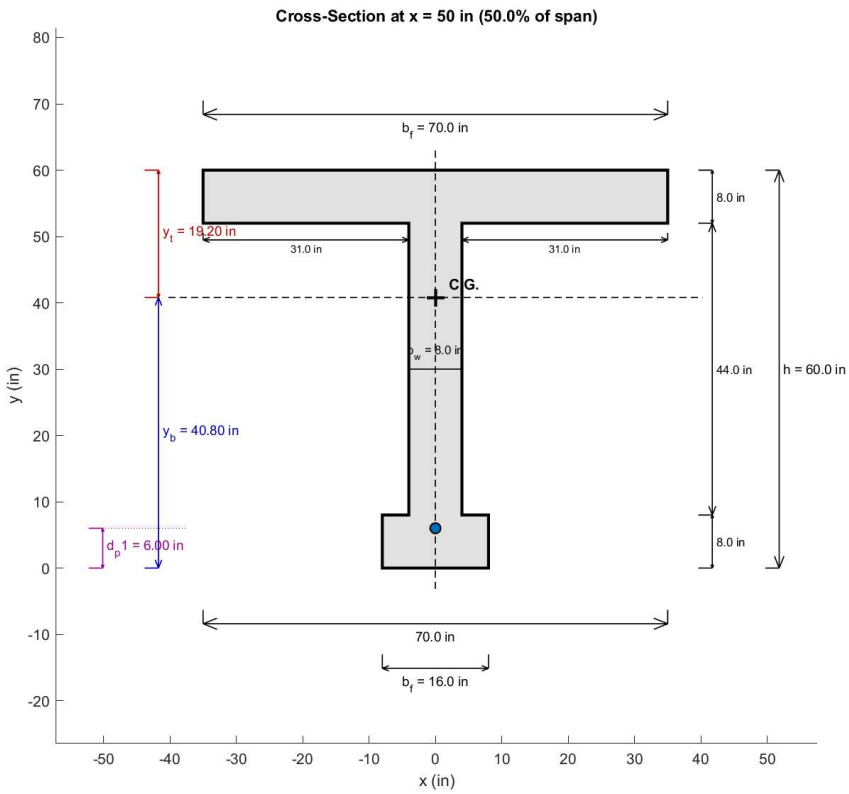
- Top fiber

$$\frac{Pey}{I} = \frac{400(26.1)(19.2)}{404241.07} = 0.496 \text{ ksi}$$

- Bottom Fiber

$$\frac{Pey}{I} = \frac{400(26.1)(40.8)}{404241.07} = -1.054 \text{ ksi}$$

At x = 0.5L,



- Top fiber

$$\frac{Pey}{I} = \frac{400(34.8)(19.2)}{404241.07} = 0.661 \text{ ksi}$$

- Bottom Fiber

$$\frac{Pey}{I} = \frac{400(34.8)(40.8)}{404241.07} = -1.405 \text{ ksi}$$

III. Total Stresses in Beam

At end of support, x = L and x = 0

At Top Fiber, σ top = $-\frac{P}{A} + \frac{PeYt}{I} - \frac{MYt}{I}$

$$\sigma = -\frac{400}{1040} + \frac{400(0)(19.2)}{404241.07} - \frac{0\ x\ 19.2}{404{,}241.07\ x\ 1000}$$

$$= -0.385 + 0 + 0$$

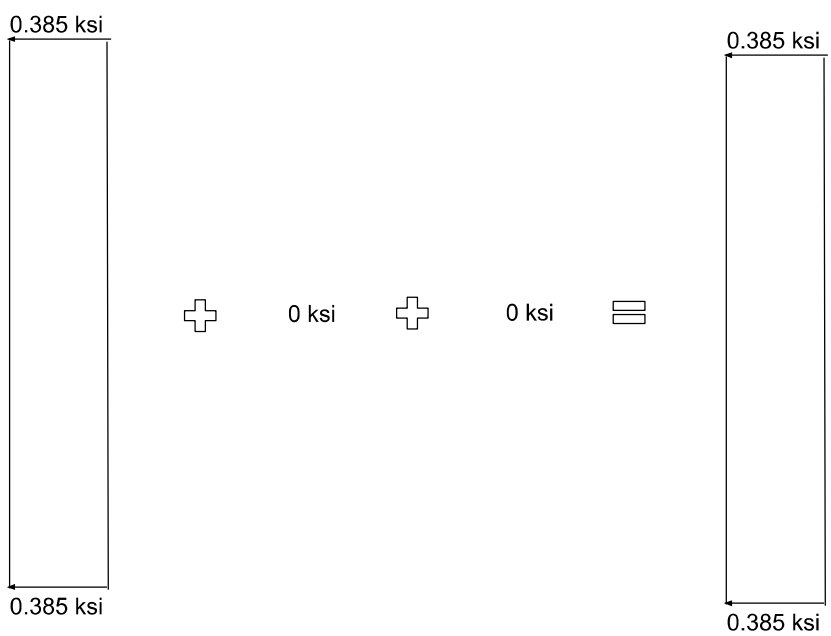
$$= -0.385 \text{ ksi (compression)}$$

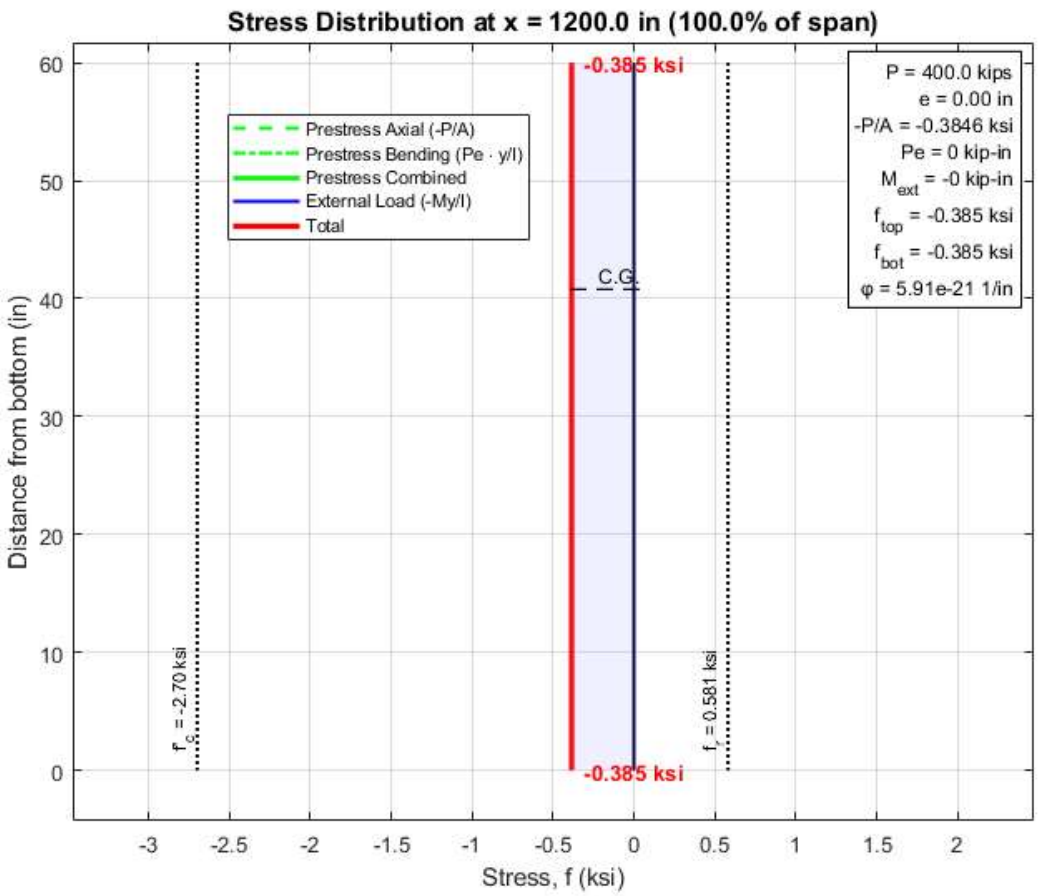
At Bottom Fiber, σ bot = $-\frac{P}{A} - \frac{PeYb}{I} + \frac{MYb}{I}$

$$\sigma = -\frac{400}{1040} + \frac{400(0)(40.8)}{404241.07} - \frac{0\ x\ 40.8}{404{,}241.07\ x\ 1000}$$

$$= -0.385 + 0 + 0$$

$$= -0.385 \text{ ksi (compression)}$$




Stress Distribution at x = 1200.0 in (100.0% of span)
Prestress Axial (-P/A)
Prestress Bending (Pe · y/I)
Prestress Combined
External Load (-My/I)
Total
P = 400.0 kips
e = 0.00 in
-P/A = -0.3846 ksi
Pe = 0 kip-in
M_ext = -0 kip-in
f_top = -0.385 ksi
f_bot = -0.385 ksi
φ = 5.91e-21 1/in
-0.385 ksi
-0.385 ksi
C.G.
f'_c = -2.70 ksi
f_r = 0.581 ksi
Distance from bottom (in)
Stress, f (ksi)

At x = 0.75L and x = 0.25L,

At Top Fiber, σ top = $-\frac{P}{A} + \frac{PeYt}{I} - \frac{MYt}{I}$

$$\sigma = -\frac{400}{1040} + \frac{400(26.1)(19.2)}{404241.07} - \frac{12188160\ x\ 19.2}{404{,}241.07\ x\ 1000}$$

= - 0.385 + 0.496 - 0.579

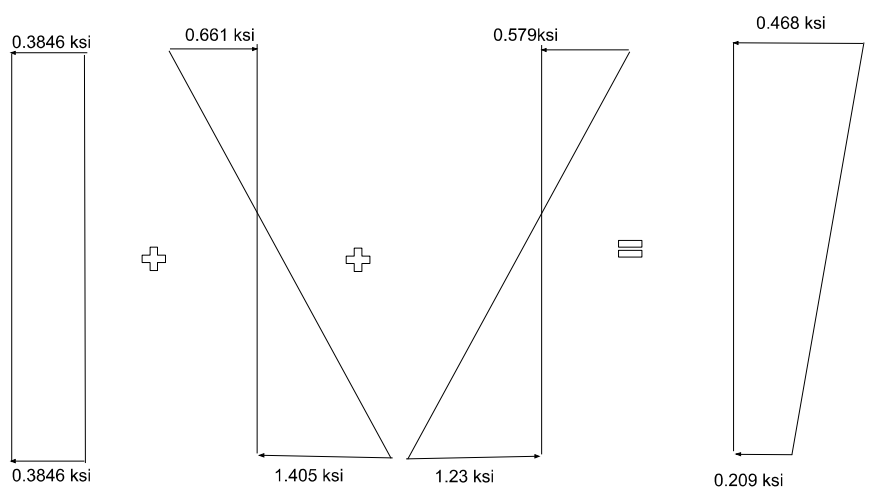
= - 0.468 ksi (compression)

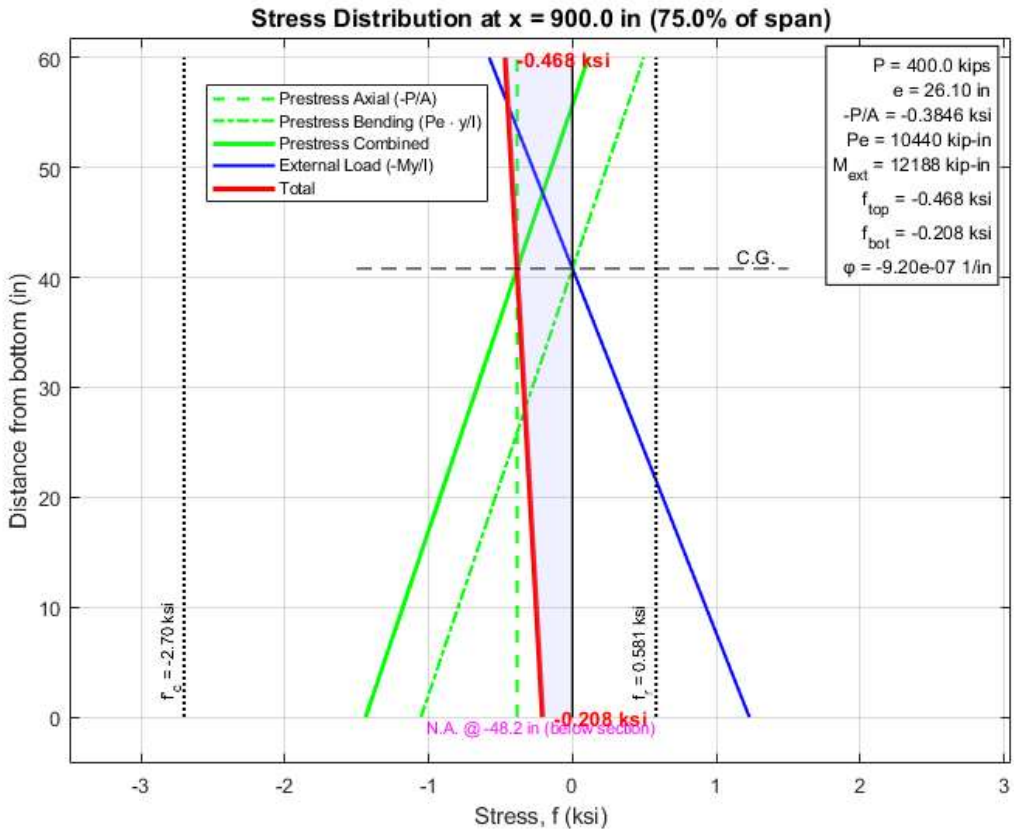
At Bottom Fiber, σ bot = $-\frac{P}{A} - \frac{PeYb}{I} + \frac{MYb}{I}$

$$\sigma = -\frac{400}{1040} - \frac{400(26.1)(40.8)}{404241.07} + \frac{12188160\ x\ 40.8}{404{,}241.07\ x\ 1000}$$

= - 0.385 - 1.054 + 1.230

= - 0.209 ksi (compression)





At x = 0.5L,

At Top Fiber, σ top = $-\frac{P}{A} + \frac{PeYt}{I} - \frac{MYt}{I}$

$$\sigma = -\frac{400}{1040} + \frac{400(34.8)\ (19.2)}{404{,}241.07} - \frac{16250400\ (19.2)}{404{,}241.07\ x\ 1000}$$

= - 0.385 + 0.661 - 0.7718

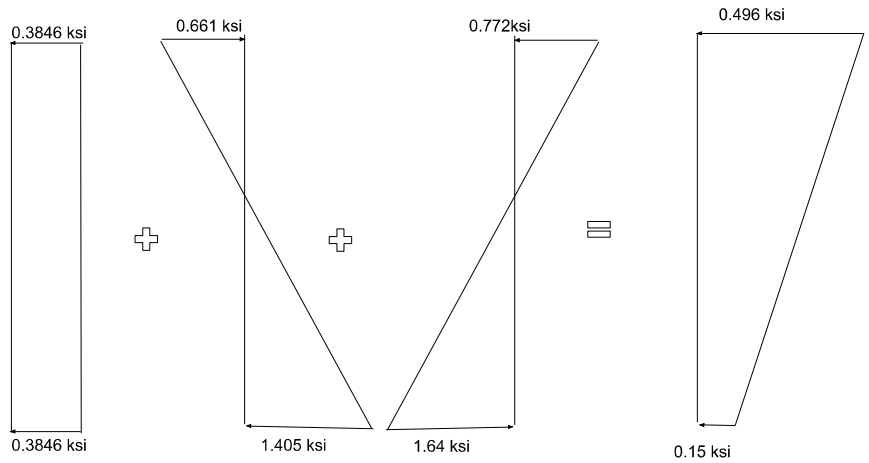
= - 0.4958 ksi (compression)

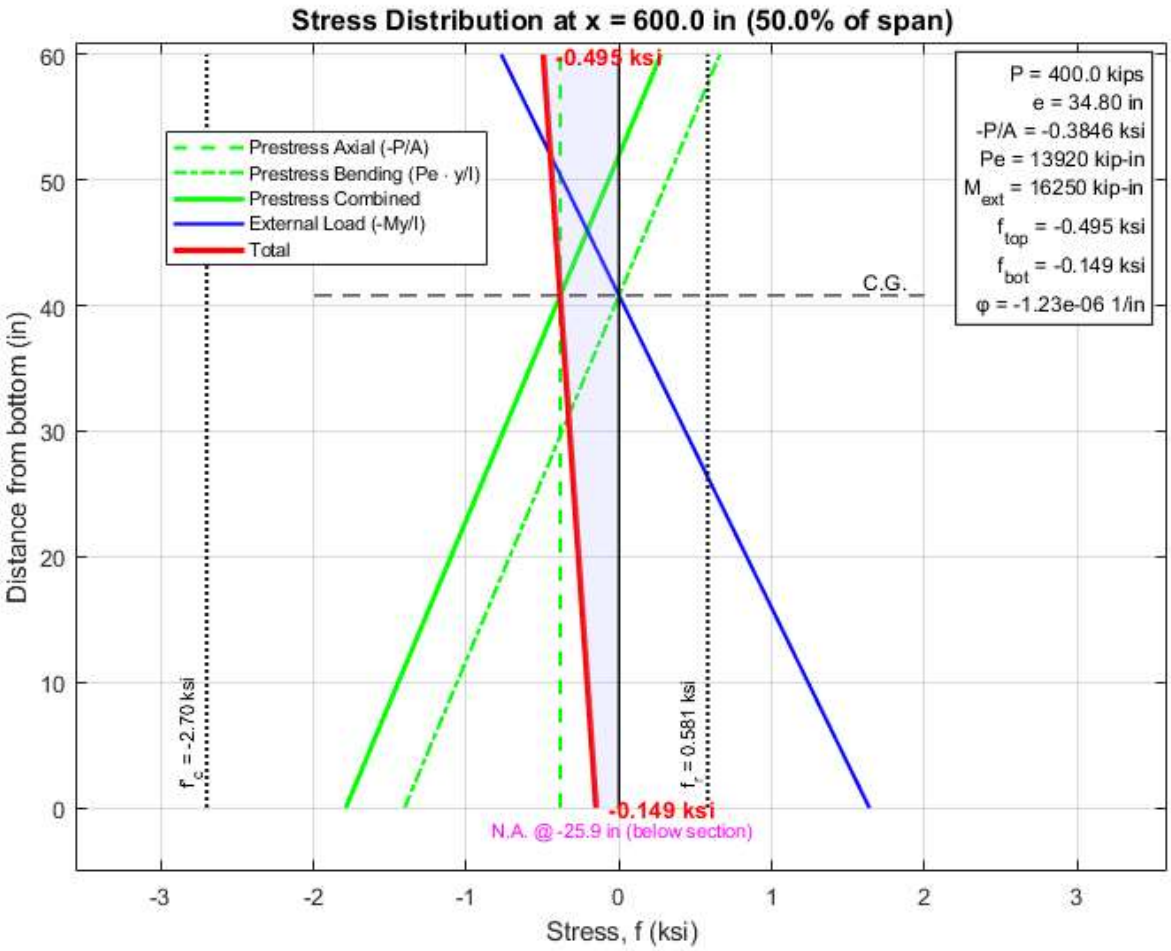
At Bottom Fiber, σ bot = $-\frac{P}{A} - \frac{PeYb}{I} + \frac{MYb}{I}$

$$\sigma = -\frac{400}{1040} - \frac{400(34.8)(40.8)}{404241.07} + \frac{16250400\ x\ 40.8}{404{,}241.07\ x\ 1000}$$

= - 0.385 - 1.405 + 1.640

= - 0.150 ksi (compression)


0.3846 ksi
0.661 ksi
0.772ksi
0.496 ksi
0.3846 ksi
1.405 ksi
1.64 ksi
0.15 ksi


Stress Distribution at x = 600.0 in (50.0% of span)
Prestress Axial (-P/A)
Prestress Bending (Pe · y/I)
Prestress Combined
External Load (-My/I)
Total
P = 400.0 kips
e = 34.80 in
-P/A = -0.3846 ksi
Pe = 13920 kip-in
M_ext = 16250 kip-in
f_top = -0.495 ksi
f_bot = -0.149 ksi
φ = -1.23e-06 1/in
-0.495 ksi
-0.149 ksi
C.G.
N.A. @ -25.9 in (below section)
f'_c = -2.70 ksi
f_r = 0.581 ksi
Distance from bottom (in)
Stress, f (ksi)